DESIGN AND EVALUATION OF AN AUGMENTED REALITY STORYBOOK FOR PRIMARY LEARNERS' PLANT GROWTH UNDERSTANDING & STEWARDSHIP

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Environmental education for young learners often relies on static teaching methods that are insufficient for developing sustained ecological responsibility. Although Augmented Reality (AR), interactive storytelling, and gamification have shown promise in educational contexts, current systems still have important limitations. Existing AR educational applications are mostly short, episodic experiences and do not support longitudinal persistence that reflects the continuous nature of biological care. In addition, there is limited use of intelligent, rule-based dialogue systems tailored to the cognitive and linguistic abilities of primary school learners, and current environmental serious games rarely include redemption-based mechanics that allow children to correct mistakes through later conservation actions. These gaps reduce the ability of existing tools to foster long-term engagement, accountability, and proactive environmental stewardship. This study proposes a gamified AR-based narrative framework that combines rule-based storytelling, time-delta persistence, intrinsic motivational design, and redemption-oriented ecological learning. The system is designed to create branching narrative experiences in which learner choices influence plant growth, mood, and story progression over time, while context-sensitive dialogue and feedback are generated through a rule-based mechanism. The framework will be developed using mobile AR technologies and interactive storytelling techniques to support meaningful learning and sustained behavioral engagement. The expected outcomes include improved narrative engagement, stronger environmental awareness, increased feelings of responsibility, and more positive prosocial ecological behavior among learners. Socially, the proposed framework can contribute to early environmental stewardship education by promoting care, empathy, and sustainable thinking from childhood. Commercially, it offers potential for adaptation into interactive educational products for schools, digital learning platforms, and child-centered environmental learning applications.

***Keywords*:** Augmented Reality, Interactive Storybook, Hybrid Interaction Framework, Grade 1 Science, Child-Centered Learning

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AR	Augmented Reality
ARCore	Android Augmented Reality Core
ARKit	Apple Augmented Reality Kit
CSEDU	International Conference on Computer Supported Education
JSON	JavaScript Object Notation
MAR	Mobile Augmented Reality
SDK	Software Development Kit
UI	User Interface

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1. INTRODUCTION

1.1 Background and Context

Environmental stewardship should begin in early childhood, because foundational learning at this stage strongly influences later attitudes, habits, and decision-making. UNESCO reports that 37% of the world's children, representing more than 300 million learners, may not achieve minimum reading proficiency by 2030 without urgent intervention, showing the wider challenge of making early learning meaningful and effective [17]. In science and environmental education, this problem is intensified when young learners are taught through static explanation rather than through interactive and experiential methods. At the same time, Sri Lanka shows growing readiness for digital learning: in early 2025, the country had 12.4 million internet users, 29.3 million mobile connections, and an internet penetration rate of 53.6%, indicating practical potential for mobile-based educational solutions [18]. Within this context, AR-supported learning becomes important because mobile augmented reality can place digital content into real-world contexts and support situated learning experiences that traditional media cannot provide [2], [3], [5]. Studies also show that AR educational games can improve both achievement and learner attitudes during real-world observation tasks [8]. However, existing solutions remain limited. Current AR educational applications are mostly short, episodic experiences and do not simulate the long-term commitment required for biological care [5], [6]. Research on pedagogical interactive narrative and intelligent interfaces demonstrates the value of adaptive storytelling [1], [3], yet child-appropriate rule-based dialogue for plant-focused learning remains underdeveloped, while overly complex mobile AR interaction can increase mental workload and cognitive fatigue [4]. Likewise, gamification literature shows that shallow reward structures often fail to support deeper intrinsic motivation and sustainable behavior change [9], [10], [12].

To address this problem, the proposed research introduces a handheld AR-based gamified narrative framework for environmental learning. The overall solution combines dynamic rule-based storytelling, time-delta narrative persistence,

intrinsically meaningful gamification, and redemption-based ecological learning so that children can observe how their actions affect plant growth, mood, and story progression over time [1], [5], [6], [13],[16]. This direction aligns with SDG 4: Quality Education and SDG 13: Climate Action, as it supports engaging, technology-enhanced science learning while also promoting environmental responsibility [19], [20]. It also fits the Mixed Reality / Interactive Media research cluster because it integrates AR interaction, intelligent narrative design, behavioral motivation, and child-centered digital learning. Within this broader problem, the present component specifically focuses on developing the gamified narrative engine, particularly the use of rule-based adaptive storytelling, narrative persistence, and redemption-oriented feedback to foster sustained ecological responsibility in young learners.

1.2 Literature Review

Recent research shows that the combination of augmented reality, interactive storytelling, intelligent interfaces, and gamification has strong potential for educational use. Interactive narrative helps learners become active participants rather than passive receivers of information, while theory-of-mind-based character design allows digital systems to respond in more pedagogically meaningful ways [1]. Mobile augmented reality further strengthens this by connecting virtual content with the real environment, creating more immersive and context-sensitive learning experiences [2], [3]. Studies on mobile educational AR games also show that such systems can improve engagement and support experiential learning when they are designed as interactive environments rather than simple visual tools [5].

The literature also suggests that AR learning becomes more effective when systems support continuity, responsiveness, and meaningful learner participation. AR simulation and game-based studies show that virtual entities can encourage observation, care, and long-term interaction, helping learners develop stronger involvement and responsibility [6], [7]. In addition, AR-based educational games have been found to improve both learning achievement and learner attitudes during real-world observation tasks [8]. However, important challenges remain. Non-linear storytelling is still difficult to design and maintain, especially when narrative coherence must be preserved while adapting to user choices [2]. Mobile AR systems can also increase usability problems, mental workload, and interaction complexity if they are not carefully designed [3], [4].

Gamification adds further value by helping sustain motivation and participation, but its effectiveness depends on how meaningfully it is integrated with learning goals [9]. Therefore, current literature supports the development of an educational AR system that combines storytelling, intelligent interaction, simulation, and motivation as one integrated learning experience rather than as separate features. Together, these studies provide a clear foundation for designing AR-based educational environments that are immersive, engaging, and pedagogically purposeful [1], [9].

1.2.1 Dynamic Narrative Generation using Rule-Based AI

Dynamic narrative generation has become an important part of modern educational technology because it transforms learners from passive observers into active participants within an evolving digital story [1]. In educational settings, this is especially valuable because it allows learners to interact with content in a more meaningful and engaging way. Instead of simply receiving information, learners can influence events, make choices, and experience the consequences of their actions within the learning environment.

The integration of Interactive Storytelling (IS) with Mobile Augmented Reality (MAR) has further strengthened this approach by creating immersive learning experiences that traditional media cannot provide. Through MAR, digital narrative elements can be placed within the learner's physical environment, making the experience more contextual and realistic. However, this combination also introduces major design challenges. In particular, authoring non-linear narratives and maintaining narrative coherence in a 3D augmented space are technically complex tasks [2].

To address these challenges, researchers have moved away from simple static structures such as fixed linear paths or basic decision trees and have instead focused on more advanced rule-based architectures. One important development is the use of autonomous agents that encode a Theory of Mind, allowing digital characters and systems to make decisions based on beliefs, intentions, motivations, and social expectations [1]. This enables the system to adapt to learner input while still preserving pedagogical value. Such intelligent responsiveness is also related to the concept of Pose-Interfaced Presentation (PIP)**,** where the interface responds to the learner's gestures, intentions, and spatial position within the AR environment [3].

Despite these advancements, dynamic narratives must be designed carefully to avoid creating excessive mental demand on the user. Studies show that unpredictable or highly complex narrative changes can increase mental workload and lead to cognitive fatigue, especially in mobile AR settings [4]. Therefore, an effective rule-based AI engine should not only generate branching story paths, but should also do so in a way that reduces cognitive disorientation and supports concentration and immersion [3], [4]. By using rapid prototyping methods and established narrative templates, developers can create more robust and educationally suitable MAR storytelling systems [2].

1.2.2 The Intersection of Narrative Persistence and AR Learning

The integration of narrative persistence within Augmented Reality (AR) learning environments represents a significant shift toward immersive biological simulation and longitudinal engagement. Mobile AR is defined as a technology that displays and interacts with virtual content overlaid on a real-time camera feed, providing significant affordances for context-sensitive learning and situated cognition [5].

Within this framework, narrative persistence specifically the ability of a virtual entity to exist, evolve, and progress over time independent of active user sessions is best exemplified in virtual pet simulators. Research into AR-based simulators demonstrates that these systems allow users to perform continuous care tasks, such as feeding, watering, or monitoring health, which directly mirrors the responsibility and daily commitment required in real-world environmental stewardship [6].

Furthermore, AR games have proven exceptionally effective in modeling complex conceptual processes that occur over long durations, such as biological life cycles, symbiotic relationships, and evolutionary strategies, enabling learners to visualize and manipulate processes that are otherwise intangible or invisible to the naked eye [7]. This persistence is critical for educational games aiming to move beyond simple knowledge acquisition to change learner attitudes; empirical evidence suggests that AR-based educational games significantly improve students' learning achievements and their positive attitudes toward real-world observations compared to traditional classroom methods [8].

By leveraging time-delta-based persistence mechanisms, AR frameworks can successfully bridge the temporal gap between digital interaction and real-world biological growth, ensuring that the narrative continues to unfold and respond to the "passage of time" even when the user is not actively engaged with the device [5], [6]. Thus, the intersection of persistence and AR learning provides a robust foundation for fostering a long-term sense of accountability and environmental insight in primary learners.

1.2.3 Critical Evaluation of Behavioral Gamification Models

Behavioral gamification models involve the strategic and systemic use of game design elements in non-game contexts to drive deep engagement, intrinsic motivation, and sustainable behavior change [9]. These models are predominantly grounded in Self-Determination Theory (SDT), which identifies autonomy, competence, and relatedness as the three core psychological needs required to foster genuine intrinsic motivation rather than temporary compliance [10], [11].

While gamification is often implemented globally to improve learner performance, a critical evaluation of current literature reveals significant organizational and design-related impediments that can hinder its long-term effectiveness, particularly in educational settings [9]. A primary criticism involves the pervasive over-reliance on "shallow" gamification mechanics such as points, badges, and leaderboards (PBL) which focus almost exclusively on extrinsic motivation.

Research suggestively argues that while these extrinsic rewards may provide immediate engagement, they often fail to internalize the target behavior and can even diminish existing intrinsic interest if the rewards are perceived as controlling or disconnected from the task [10], [11]. Furthermore, the interaction between gamification and emergent technologies like Augmented Reality (AR) presents complex and sometimes contradictory results; empirical evidence suggests that while AR significantly enhances motivation compared to conventional materials, the additive layer of traditional gamification elements does not always yield a statistically significant increase in motivation or learning outcomes over AR alone [12].

This indicates that for behavioral models to be effective, gamification should move beyond superficial rewards toward meaningful design that facilitates active user participation and provides a sense of redemption or narrative progress that is intrinsically and pedagogically tied to the content [9], [12]. Thus, a robust behavioral framework must prioritize the psychological impact of game mechanics on user autonomy to ensure long-term stewardship and successful behavior modification [10], [11].

1.2.4 Digital Storytelling and Prosocial Behavior in Early Childhood

Digital storytelling serves as a powerful pedagogical tool for cultivating environmental literacy and prosocial behavior during early childhood, a period widely identified as critical for developing lifelong social, moral, and ethical values [13]. Interactive digital technologies, including Augmented Reality (AR) and mobile applications, can significantly pique student interest and engage them in active environmental stewardship by enabling them to capture, document, and share experiences of local ecosystems in real time [14].

Research indicates that integrating ecology-based storytelling into the primary school curriculum provides a comprehensive framework for promoting sustainable living and environmental management from a young age, effectively habituating care for the planet [15]. Furthermore, AR solutions demonstrate significant potential for behavioral learning, particularly in improving socio-emotional functioning by providing immersive coaching and immediate narrative feedback based on the child's interactions [16].

By moving beyond static instruction toward experiential, hands-on approaches, digital storytelling allows children to witness the direct, often irreversible consequences of their actions on virtual environments, thereby fostering a deep-seated sense of responsibility and empathy for nature [13], [14]. This transformation of learning through innovative digital narratives is essential for strengthening the character of future generations and ensuring they understand the complex interdependence of human decisions and ecological health [15].

Consequently, the convergence of AR and storytelling provides a robust mechanism for encouraging proactive, prosocial environmental behaviors in primary school-aged students, turning abstract environmental concepts into tangible, personal experiences [13], [16].

1.3 Research Gap

Overall research problem: The overall research problem of this study is that young children often find it difficult to understand plant growth, environmental changes, and ecological concepts through traditional teaching methods such as textbooks and verbal explanations alone. Although Augmented Reality has improved interactive learning, existing educational AR systems still do not fully support child-centered, emotionally engaging, and biologically meaningful learning experiences for early primary learners.

My component specific research problem: The research problem is how to design a gamified narrative framework for a handheld AR storybook that can make children's plant-care choices meaningful over time through rule-based adaptive storytelling, persistent virtual plant behavior, non-point ecological rewards, and redemption-based corrective actions, so that environmental responsibility and feelings of care can be improved in young learners.

Augmented Reality and gamification have increasingly been explored as promising approaches for improving educational engagement, immersion, and learner participation. Prior research shows that interactive pedagogical narratives can support more meaningful learner involvement by allowing digital characters and systems to respond to user behavior in adaptive ways [1]. Mobile Augmented Reality has also been shown to create contextual and interactive learning environments by overlaying digital content onto the real world, while AR-based educational games have demonstrated positive effects on learner achievement and attitudes [2], [3], [5], [8]. In addition, gamification research indicates that game design elements can improve participation and motivation when they are carefully aligned with learning goals [9], [10], [12]. Environmental education studies further suggest that digital technologies can support environmental awareness and encourage positive behavior in school-aged learners [13],[16]. However, although these studies provide valuable foundations, their findings remain fragmented and do not yet offer a sufficiently integrated framework for developing effective environmental stewardship tools for young learners.

This limitation is important because environmental stewardship requires more than short-term engagement or isolated learning gains. Children must be supported not only to understand environmental concepts, but also to experience continuity of care, age-appropriate guidance, meaningful motivation, and opportunities to correct harmful choices. Existing systems frequently succeed in one area, such as immersive visualization, gamified participation, or adaptive interaction, but fail to combine these elements into a pedagogically coherent learning experience. As a result, the literature still lacks a strong framework capable of sustaining long-term ecological responsibility in young learners through an AR-based educational environment [5], [8], [9], [13], [16].

Identified Research Gap 1 – Lack of longitudinal persistence in AR educational systems: Existing AR learning applications improve engagement and observation, while AR simulators show that virtual entities can support repeated care tasks, but most systems still remain one-time and session-based rather than evolving over time, limiting long-term responsibility and biological care learning [5],[8].

The first gap concerns the lack of longitudinal persistence in existing AR educational systems. Most AR learning applications are designed as short, episodic experiences that begin and end within a single user session [5], [8]. While this approach can improve immediate engagement and support observation-based learning, it does not reflect the continuous and cumulative nature of biological care. Virtual pet simulators demonstrate that AR can support repeated care actions over time, such as monitoring, feeding, and maintaining the condition of a digital entity [6]. Likewise, AR-based biology games show that long-duration ecological and biological processes can be represented through digital interaction [7]. However, these ideas have not been sufficiently adapted into educational environmental stewardship systems for children. Consequently, current AR learning tools remain limited in their ability to simulate the long-term commitment required for caring for living things. What is still missing is a persistence mechanism that allows a virtual living system to evolve over time, even in the learner's absence, thereby reinforcing continuity, accountability, and ecological responsibility.

Identified Research Gap 2 – Absence of child-centered rule-based dialogue systems: Existing pedagogical interactive narratives and intelligent AR interfaces support adaptive interaction, but most systems still do not provide rule-based dialogue that is cognitively manageable, pedagogically guided, and developmentally suitable for young learners, causing either rigid static feedback or excessive complexity [1],[4].

The second gap is the absence of intelligent, rule-based dialogue systems tailored to the cognitive and linguistic levels of young learners within educational AR environments**.** Existing studies on pedagogical interactive narrative and intelligent interfaces show that adaptive systems can respond to learner actions and create more meaningful story-based interaction [1], [3]. Research on mobile AR storytelling also confirms the feasibility of non-linear narrative applications for learning [2]. However, these studies do not sufficiently address how such adaptive narrative systems can be designed for children in ways that remain pedagogically safe, understandable, and developmentally appropriate. Many existing systems either depend on rigid static text, which limits responsiveness, or on more complex adaptive mechanisms that may lack adequate pedagogical control for primary education. This problem becomes more significant because evidence shows that mobile AR storytelling can increase mental workload and usability challenges if interaction complexity is not carefully managed [4]. Therefore, the literature still lacks a child-centered rule-based dialogue framework that can provide coherent, context-sensitive, and educationally guided responses suitable for young learners in plant-based environmental learning contexts.

Identified Research Gap 3 – Limited behavioral gamification for intrinsic ecological responsibility: Existing gamification research shows that game elements can enhance motivation and participation, but most educational systems still rely on external reward structures rather than meaningful ecological consequence and learner agency, resulting in short-term engagement instead of sustained stewardship behavior [9], [10], [12], [13].

The third gap concerns the limited use of meaningful behavioral gamification models for environmental stewardship learning. Current research confirms that gamification can improve learner motivation, engagement, and participation [9], [10]. However, much of this work relies heavily on traditional reward systems such as points, badges, and external progress indicators. Even in AR-supported learning, evidence suggests that although AR itself can significantly increase motivation, the additional use of conventional gamification elements does not always result in substantially better learning outcomes [12]. This suggests that many existing gamification approaches are too superficial to support deeper, long-term behavior change. In environmental education, digital games and serious games have successfully raised awareness and participation among children [13], [14], but they often focus on short-term involvement rather than intrinsic responsibility. Therefore, current approaches remain insufficient because they do not adequately explain how gamification can be structured to foster internalized ecological care rather than temporary compliance. A more robust behavioral model is needed, one that connects motivation directly to consequence, responsibility, and learner agency.

Identified Research Gap 4 – Limited use of redemption mechanics in environmental learning systems: Existing environmental educational games and AR behavioral learning systems highlight ecological consequences, but few provide structured opportunities for children to actively correct mistakes through later conservation actions, reducing the potential for proactive and restorative environmental learning [13],[16].

The fourth gap relates to the limited use of redemption mechanics in environmental educational games and AR learning systems. Existing environmental learning tools often highlight the negative consequences of harmful actions or ecological neglect [13],[16]. They can teach learners that poor choices damage virtual environments or lead to undesirable outcomes. However, few systems provide a structured narrative framework through which children can actively correct those mistakes through later conservation-oriented actions. This is a major limitation because environmental learning should not only teach consequence, but also encourage constructive recovery and proactive responsibility. Without a redemption mechanism, children may learn that harmful actions produce damage, but they may not learn that responsible action can restore or improve the situation. These risks creating a more passive or defeatist learning experience rather than a hopeful and action-oriented one. Therefore, there is still a clear need for an eco-pedagogical model that integrates redemption as a core mechanic, allowing children to learn stewardship through consequence, correction, and restoration.

Taken together, these gaps justify the need for a new framework that combines longitudinal persistence, child-centered rule-based dialogue, meaningful gamification, and redemption-based ecological learning within one AR-supported educational system. Existing studies provide strong evidence for the individual value of AR interaction [2], [5], [8], adaptive narrative [1], [3], gamification [9], [10], [12], and environmental digital learning [13],[16], but they do not yet combine these elements in a way that supports sustained engagement and long-term ecological stewardship among young learners. The proposed study therefore addresses a clear evidence-based gap by moving beyond isolated technological features toward a more integrated and pedagogically grounded model for environmental responsibility learning.

Table 1.1: Research Gap Comparison

Gap	What Existing Studies Address	What is Still Missing	Proposed Contribution
Persistence in AR learning	AR games improve engagement, and some AR simulators model ongoing care and biological processes [5]–[8]	Most systems are still short-term and do not support continuous plant-care learning over time	A time-delta persistence model for plant growth, decay, and recovery across sessions
Rule-based child dialogue	Adaptive storytelling and intelligent interfaces support responsive AR narratives [1]–[3]	No child-centered rule-based dialogue system for plant-focused AR learning; complex systems may increase mental load [4]	A child-appropriate rule-based narrative engine with clear, guided dialogue
Meaningful gamification	Gamification improves motivation and participation in education [9], [10], [12]	Most systems still rely on external rewards rather than intrinsic ecological responsibility	A non-point gamification model based on consequence, care, and learner agency
Redemption-based learning	Environmental games and AR learning support awareness and positive behavior [13], [14], [16]	Few systems let children correct mistakes through later conservation actions	A redemption-oriented eco-pedagogical model based on recovery and restoration

2. OBJECTIVE

2.1 Main Objective

The main objective of this study is to design, develop, and evaluate a gamified AR-based narrative framework that uses rule-based intelligent storytelling, time-based narrative persistence, intrinsic motivational design, and prosocial environmental feedback to promote sustained ecological responsibility among primary school learners**.** This objective will be achieved by integrating four core elements into a single educational experience: a rule-based branching narrative that adapts to learner choices, a persistent virtual plant system that changes over time, a non-point motivational structure that emphasizes intrinsic engagement rather than external rewards, and a digital storytelling approach that encourages empathy, responsibility, and prosocial environmental behavior. Prior research shows that pedagogical interactive narrative can make learners active participants rather than passive observers [1], while mobile augmented reality can provide immersive, context-sensitive educational experiences that traditional media cannot replicate [2], [3], [5]. Existing studies also indicate that persistent AR simulations can model long-term care and biological change [6], [7], that gamification can support motivation when meaningfully aligned with learning goals [9], [10], [12], and that digital storytelling can strengthen environmental literacy and prosocial behavior in children [13],[16]. The effectiveness of this objective will be measured through narrative coherence, persistence accuracy, engagement with non-point motivational mechanics, and changes in environmental attitudes and feelings of responsibility observed through pre- and post-interaction evaluation.

2.2 Specific Objectives

To achieve the main objective, the study will pursue the following specific objectives. These objectives divide the research into clear and measurable parts so that the proposed narrative framework can be systematically designed, implemented, and evaluated. Each objective is directly aligned with the identified research problem and with the major dimensions of the component, namely intelligent storytelling, persistence, motivational design, and prosocial environmental learning.

Table 2.1: Specific Objectives of the Research

Number	Objective
1	To design and evaluate a **rule-based branching narrative system** that generates context-sensitive dialogue and story progression according to learner choices, plant condition, and environmental context.
2	To develop and test a **time-delta-based persistence mechanism** that enables the virtual plant to grow, decay, or recover across inactive periods, thereby representing long-term biological change within the AR learning experience.
3	To design and assess a **non-point gamification model** that promotes intrinsic motivation through narrative consequence, ecological meaning, and learner agency rather than through points, badges, or leaderboards.
4	To develop and evaluate a **redemption-based storytelling structure** that allows learners to correct harmful choices through later conservation actions and thereby supports restorative ecological learning.

3. METHODOLOGY

This study will adopt a Design Science Research approach supported by prototype-based experimental evaluation. This approach is appropriate because the study aims to design, implement, and validate a functional pedagogical narrative component for a handheld AR learning system rather than merely describe an existing phenomenon. The proposed work focuses on developing and testing a mobile AR narrative framework that integrates rule-based storytelling, time-based persistence, meaningful gamification, and prosocial environmental learning. Since prior research highlights both the educational potential and the design complexity of combining interactive storytelling, augmented reality, and gamification [1], [5], the methodology is structured to support both artifact development and empirical evaluation. The methodology is structured around four specific objectives. The detailed methodology for each objective is summarized in Tables 3.1 to 3.4.

Table 3.1: Methodology for Dynamic Narrative Generation using Rule-Based AI

Aspect	**Methodology**
Research approach	A design-and-evaluate approach will be used to create a rule-based narrative engine for a mobile AR learning experience.
Technical methods and algorithms	The system will use a branching narrative graph in which nodes represent story events, dialogue states, and ecological situations, while branches represent learner choices. A rule-based decision engine will determine the next dialogue and event according to variables such as prior choices, plant condition, and current narrative state. This is suitable because prior work shows that pedagogical interactive narrative and theory-of-mind-based character design can support adaptive educational storytelling [1].
Tools and platforms with justification	Unity 3D and C# will be used to implement dialogue logic, event management, and AR interaction flow. AR Foundation will support deployment on mobile AR platforms. These tools are appropriate because they support rapid prototyping of interactive storytelling and mobile AR applications [2].

Data collection and ethical considerations	Data will include dialogue-trigger logs, branch choices, session flow records, and observation notes on learner understanding and response. Because the study involves children, all data will be anonymized, participation will require parental consent and child assent, and the narrative content will remain age-appropriate and non-distressing.
Validation strategy and performance metrics	Validation will assess narrative coherence, dialogue relevance, branch completion rate, consistency between learner action and system response, and observed story comprehension. Since prior studies note that complex dynamic narratives can increase cognitive strain [4], evaluation will also examine whether the dialogue remains understandable and manageable for learners.

Table 3.2: Methodology for The Intersection of Narrative Persistence and AR Learning

Aspect	**Methodology**
Research approach	An artifact-development and experimental validation approach will be used to design a persistent AR environment in which a virtual plant changes across time.
Technical methods and algorithms	A time-delta persistence mechanism will be implemented. The virtual plant will be modeled using state variables such as growth, health, hydration, and recovery status. When the learner returns to the application, elapsed time since the previous session will be calculated and used to update the plant state. This is consistent with the persistence model discussed in AR simulators and care-based virtual systems [5], [6].
Tools and platforms with justification	Unity 3D will be used for plant-state control and AR scene management. Local persistent storage such as JSON or serialized save data will be used to store timestamps and plant-state history, because this is lightweight and sufficient for a prototype-scale mobile application.
Data collection and ethical considerations	Data will include timestamps, plant-state transitions, learner return frequency, and persistence-related system logs.

	Observation will also be used to determine whether learners understand that the virtual plant changes over time. All logs will be anonymized and used only for research purposes.
Validation strategy and performance metrics	Validation will measure state-update correctness, consistency of plant behavior across sessions, continuity of growth/decay, and learner recognition of time-based consequences. This is important because persistence is expected to support accountability and long-term environmental insight [6],[8].

Table 3.3: Methodology for Critical Evaluation of Behavioral Gamification Models

Aspect	**Methodology**
Research approach	A behavior-informed design approach will be used to develop and evaluate motivational mechanics that support intrinsic engagement rather than simple reward accumulation.
Technical methods and algorithms	The framework will use a non-point gamification model. Instead of points, badges, and leaderboards, learner motivation will be supported through narrative progress, improved plant condition, emotional feedback, and access to restorative actions. This is aligned with the literature arguing that meaningful gamification should move beyond shallow extrinsic rewards [9],[12].
Tools and platforms with justification	Unity UI, conditional event triggers, plant-state animations, and dialogue-based feedback will be used to communicate progress and motivation through story consequence rather than abstract scoring. This is suitable because AR itself can already enhance motivation, while traditional gamification layers do not always improve learning outcomes further [12].
Data collection and ethical considerations	Data will include voluntary care actions, repeat engagement, time spent on optional narrative-care tasks, and observed reactions to motivational feedback. Since the users are children, the motivational design will avoid pressure-based or manipulative reward strategies.

Validation strategy and performance metrics	Validation will focus on engagement duration, frequency of repeated care actions, completion of optional narrative tasks, and evidence of sustained participation without external score-based rewards. The subsection will be considered effective if learners stay engaged through ecological meaning and story consequence rather than through surface-level rewards alone.

Table 3.4: Methodology for Digital Storytelling and Prosocial Behavior in Early Childhood

Aspect	**Methodology**
Research approach	A pedagogical design-and-evaluation approach will be used to examine whether digital storytelling can strengthen environmental literacy, responsibility, and prosocial ecological behavior in young learners.
Technical methods and algorithms	The system will use a redemption-oriented narrative structure. If learners make neglectful or harmful choices, the plant condition and story progression will reflect those consequences. However, the system will also generate a structured opportunity for later correction through conservation actions. This supports a restorative learning model rather than a purely punitive one.
Tools and platforms with justification	Unity-based conditional branching, AR scene events, dialogue prompts, and plant-state-linked story events will be used to implement consequence and recovery. This is appropriate because AR solutions have shown potential for behavioral learning and immediate feedback [16], while digital storytelling supports prosocial and environmental value formation [13],[15].
Data collection and ethical considerations	Data will include the number of harmful choices, redemption opportunities triggered, successful restorative actions, and pre-/post-interaction responses related to environmental attitudes and responsibility. Because the work involves children, all narrative consequences will remain emotionally safe, supportive, and age-appropriate.
Validation strategy and performance metrics	Validation will assess redemption-path completion, frequency of restorative action, change in environmental attitudes, and change in responsibility-related responses between pre- and post-interaction measures. The subsection will be successful if learners not only recognize ecological consequence but also demonstrate understanding of restoration and care.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution and Its Components

The proposed solution is a handheld Augmented Reality storybook system that combines printed storybook pages, real-time camera input, mobile sensor data, and learner interaction to support plant-growth learning through an interactive and emotionally meaningful experience. The system is designed to run on standard AR-enabled smartphones and tablets, allowing digital content to be layered over physical storybook pages rather than relying on costly or physically demanding immersive hardware. Its inputs include book pages, camera feed, sensor data such as light, gyroscope, and accelerometer values, and user interactions such as touch, gestures, and device-based orientation changes. The system then processes these inputs through four coordinated modules and produces outputs such as the AR storybook experience, learning analytics, and attitude-related indicators.

At the system level, the architecture is organized into four major components. The Hybrid Interaction Framework handles tracking, physical-page anchoring, signposting, and occlusion-aware interaction. The Affective Visualization Model converts plant states into emotional visual and audio cues. The Pedagogical Narrative Engine manages branching storylines, time-based persistence, and feedback logic. The Context-Aware Simulation Engine maps environmental sensor data into plant-growth behavior and other dynamic changes. These components work together inside the process layer to transform physical interaction and environmental input into an integrated educational AR experience.

4.2 System Diagram of Component

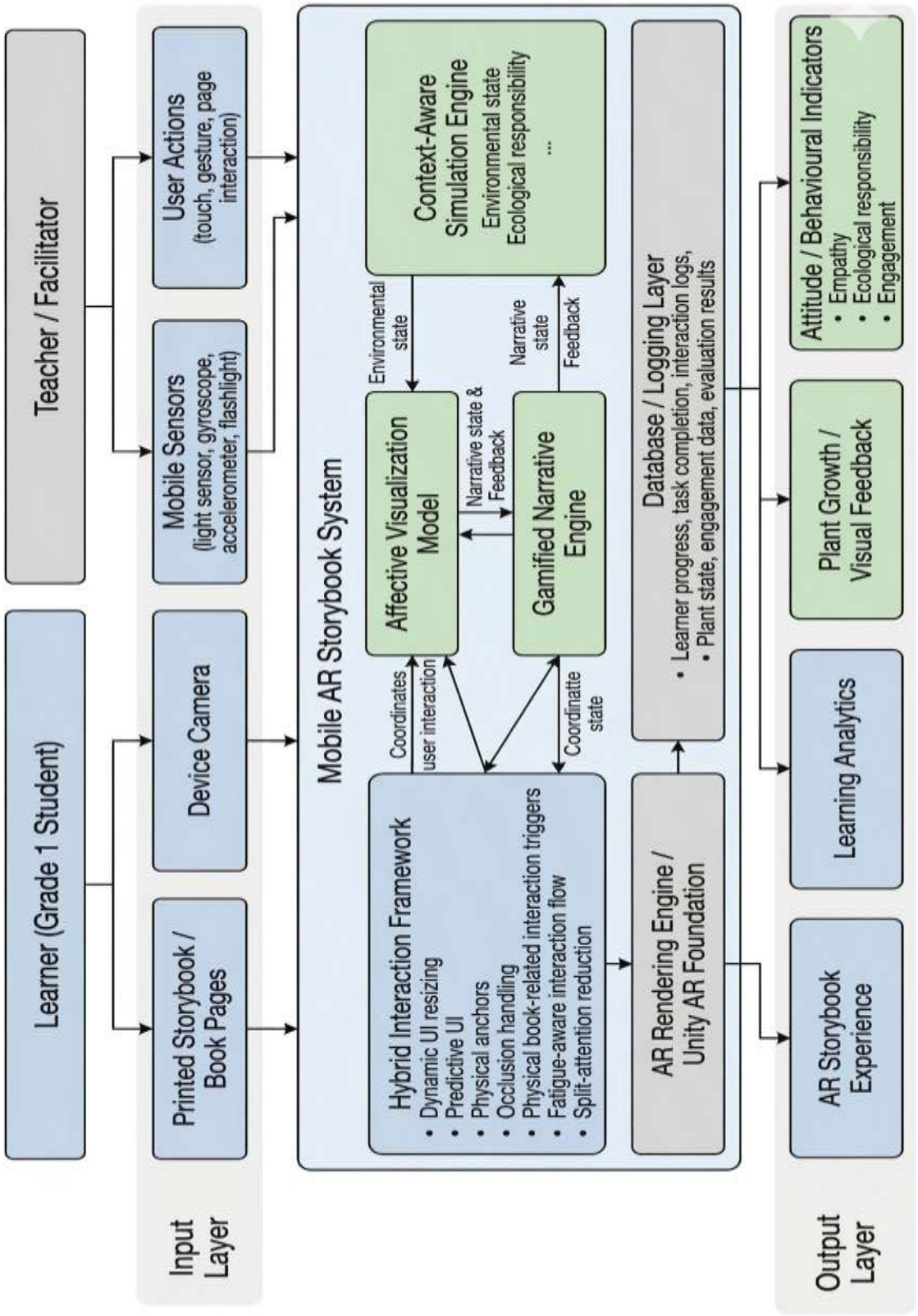


Figure 4.1: System Diagram for Components

4.3 Individual Component and Its Parts

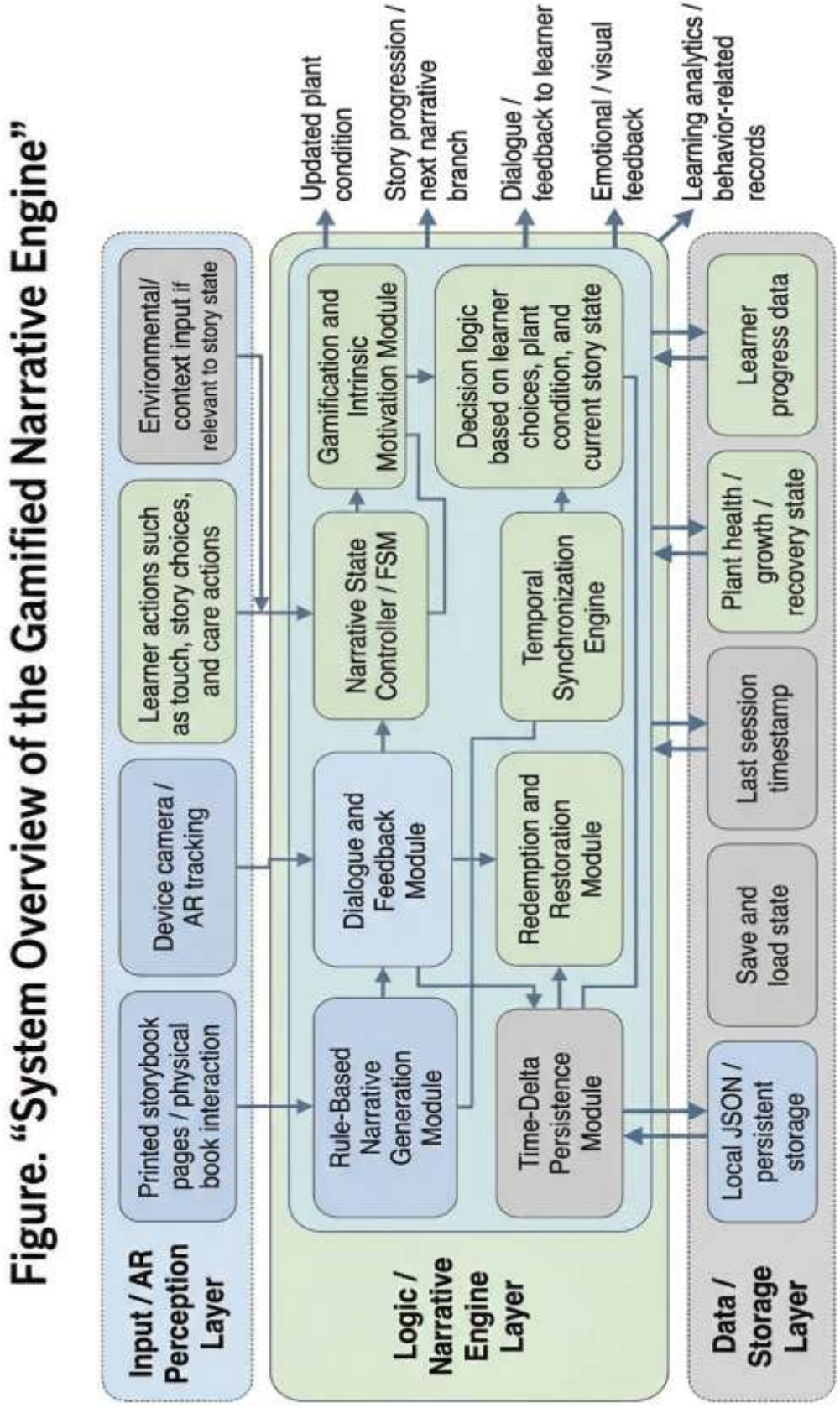


Figure 4.2: Individual Component and Its Parts

4.4 Component Interactions

The Pedagogical Narrative Engine does not operate in isolation. It receives input from the Hybrid Interaction Framework, particularly learner actions, touch events, gesture-based triggers, and physical-page interactions. These inputs determine how the narrative should progress and which story branch or dialogue response should be activated. At the same time, the narrative engine exchanges data with the Context-Aware Simulation Engine, because plant growth, stress, or recovery states generated through environmental simulation directly affect narrative events and feedback.

The narrative engine also interacts closely with the Affective Visualization Model. Once a story event or plant-state consequence has been determined, the visualization component translates that narrative meaning into visible emotional cues, such as healthy or stressed plant appearance, recovery animation, or emotionally expressive audio-visual feedback. In this way, the narrative module acts as the decision-making and consequence-management core, while the other components help collect the required inputs and deliver the resulting outputs in a learner-facing form. The process layer therefore operates as a tightly integrated environment in which interaction, simulation, visualization, and narrative continuously inform one another.

4.5 Data Flow

The data flow of the full system begins at the input layer. Printed storybook pages provide the physical anchor for AR interaction, while the camera feed identifies the page target and the mobile sensors capture environmental and device-related signals. User actions such as touch, gestures, and page interaction are then passed into the process layer.

Within the process layer, the Pedagogical Narrative Engine receives learner-choice data from the interaction module and plant-state data from the simulation module. The rule-based narrative logic then checks the current story state, plant condition, and context variables before selecting the next narrative branch, dialogue response, and consequence. If the learner has made helpful choices, the narrative may reward progress through positive plant feedback and story advancement. If the learner has made harmful or neglectful choices, the engine may trigger decline states, altered dialogue, and

redemption opportunities. The updated story state is then sent to the visualization module for learner-facing representation and can also be stored as part of learning analytics and attitude-related evaluation records.

4.6 Scalability and Security Considerations

From a scalability perspective, the proposed architecture is modular, which makes it suitable for future extension. The narrative engine can be expanded to include additional branches, dialogue sets, plant conditions, ecological scenarios, and learning topics without requiring a complete redesign of the whole system. Because the component relies on structured rule-based logic and persistent state management, new storylines, conditions, and redemption paths can be added incrementally. Similarly, the broader system can be scaled by integrating more sensor-driven simulation features, more visualization states, or additional analytics outputs while preserving the same overall architecture. The modular separation of interaction, visualization, narrative, and simulation supports maintainability and later growth.

From a security and privacy perspective, the system must be designed carefully because it involves child participants and learning analytics. Project documentation already identifies observation, pre-test and post-test questionnaires, system logs, and plant-growth-status data as relevant sources of research data, and it also specifies the need for parental consent, child assent, anonymization, and institutional ethical compliance. Therefore, the architecture should ensure that learner interaction logs, narrative progression data, and attitude-related records are stored securely and handled only for research purposes. Personally identifiable information should be minimized, and any evaluation data must be anonymized before analysis. These protections are especially important because the narrative engine records sensitive behavior-related information such as learner choices, redemption attempts, and responsibility-linked outcomes.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed system involves several key stakeholders whose needs influence the design and implementation of the solution. The primary stakeholder of the proposed system is the primary school learner aged 6-8 years, because the AR learning environment and gamified narrative experience are designed to support the child's direct interaction with the system. The learner engages with the application through story-based tasks, visual prompts, care-related actions, and interactive feedback, and therefore the system must be designed according to the child's cognitive ability, attention span, and level of understanding. Teachers and educators are also important stakeholders, because they may integrate the system into learning activities, guide students during use, and evaluate its educational value and behavioral impact. The researchers and developers are another key stakeholder group, since they are responsible for designing, implementing, and testing the gamified narrative engine as well as the overall AR learning experience. In addition, parents or guardians may be considered supporting stakeholders because they are concerned with the child's safety, usability, and educational benefit, while school administrators are also relevant stakeholders due to their interest in the practicality, relevance, and educational alignment of the proposed system within the school context.

5.2 Functional Requirements

The functional requirements describe what the system must do in order to support the intended learning experience and the research objectives.

FR1. The system shall allow users to interact with a virtual plant or environmental entity through a mobile AR interface.

FR2. The system shall generate narrative events dynamically using a rule-based AI approach rather than a fully fixed linear storyline.

FR3. The system shall adapt story progression based on user actions, choices, and care-related behaviors.

FR4. The system shall maintain narrative persistence, enabling the virtual environment or entity to evolve over time even when the user is not actively engaged in a session.

FR5. The system shall provide age-appropriate dialogue, prompts, and feedback suitable for children aged 6.

FR6. The system shall include gamification elements that encourage continued engagement and responsible behavior, such as progress tracking, rewards, recovery opportunities, or meaningful story advancement.

FR7. The system shall support redemption-based interactions, allowing users to correct previous mistakes through positive in-game actions.

FR8. The system shall provide feedback on the consequences of user actions to strengthen environmental awareness and prosocial learning.

FR9. The system shall record relevant interaction data required for evaluating user engagement, narrative flow, and behavioural response.

FR10. The system shall present the narrative in a way that supports learning without overwhelming the user with excessive complexity or disorientation.

5.3 Non-Functional Requirements

The non-functional requirements define the quality attributes the system must satisfy. These are especially important because they directly affect the effectiveness of the gamified narrative engine, which is the central research component of this study.

NFR1.Usability: The system shall be simple, intuitive, and easy to use for children aged 6. Instructions, interface elements, and narrative prompts must be clear and age-appropriate. Usability directly affects how well children can understand and respond to the narrative. If the narrative engine produces interactions that are too complex, unclear, or cognitively demanding, users may become confused and lose engagement. Therefore, the engine must deliver story events, choices, and feedback in a simple and child-friendly manner.

NFR2. Performance: The system shall respond to user interactions within an acceptable time and maintain smooth AR performance during narrative progression.

The gamified narrative engine depends on timely rule evaluation and story updates. Delays in generating narrative responses can break immersion and reduce the effectiveness of the storytelling experience. Good performance is therefore necessary to ensure that narrative transitions feel immediate, natural, and engaging.

NFR3. Reliability: The system shall operate consistently without frequent crashes, loss of progress, or incorrect story transitions. Reliability is essential because the narrative engine manages persistent story states, branching events, and user progress over time. If the system fails to preserve narrative continuity, the educational and emotional value of the experience will be weakened. A reliable engine ensures that user actions lead to stable and meaningful narrative outcomes.

NFR4. Security and Data Protection: The system shall protect user data, especially any stored progress, interaction records, or research-related usage data. Although the narrative engine primarily focuses on storytelling and engagement, it may depend on stored user progress and behavioral data to maintain persistence and adaptation. Secure handling of this data is necessary to preserve trust, protect child users, and ensure that the research component is ethically implemented.

NFR5. Maintainability: The system shall be designed in a modular and structured manner so that rules, dialogue content, and story logic can be updated easily. Maintainability is particularly important for the research component because the gamified narrative engine may require continuous refinement during development and evaluation. A maintainable rule-based structure allows the researcher to modify narrative rules, adjust behavioral responses, and test alternative storytelling strategies without redesigning the entire system.

NFR6. Scalability: The system shall support future extension of story content, additional narrative rules, and more interactive scenarios. The narrative engine must be able to grow beyond a small prototype. If the system cannot support additional rules or narrative branches, its research value will be limited. Scalability ensures that the engine can later accommodate richer storylines, more educational themes, and more advanced gamification mechanics.

NFR7. Accessibility and Learnability: The system shall be easy for first-time users to learn, with minimal guidance required to begin interaction. The success of the narrative engine depends on users quickly understanding the connection between their

actions and the unfolding story. If the system is difficult to learn, users may not fully engage with the narrative logic. Therefore, the engine must present its cause-and-effect structure in a way that is immediately understandable.

NFR8. Narrative Coherence: The system shall ensure that generated story events remain logically connected and educationally meaningful. This is one of the most critical quality requirements for the research component. Since the system uses rule-based dynamic narrative generation, the engine must maintain consistency between previous actions, current story state, and future events. Strong narrative coherence ensures that the gamified experience feels believable, supports learning objectives, and avoids confusing or contradictory story outcomes.

NFR9. Cognitive Suitability: The system shall present story interactions at a level appropriate to the cognitive abilities of primary school children. The narrative engine must balance interactivity and simplicity. If the generated narrative becomes too unpredictable or mentally demanding, it may increase cognitive load and reduce concentration. Therefore, the engine must produce manageable branching scenarios that support immersion without causing fatigue or disorientation.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the proposed solution is international schools in Sri Lanka that follow the Cambridge Primary Science curriculum, especially schools that seek interactive and technology-supported learning tools for early primary education. The system is designed as a handheld AR storybook that runs on standard ARCore- and ARKit-enabled smartphones and tablets, which makes it suitable for schools that already use mobile devices or tablets in classroom activities. Although the AR storybook is developed according to the Cambridge Grade 1 Science syllabus in international schools, it is also appropriate for 6-year-old children who enjoy learning about plants, because the concept, visuals, and interaction style are intended for early learners.

Table 6.1: Primary Customer Persona

Category	**Details**
Persona Name	School Decision-Maker
Role	Principal / Sectional Head / ICT Coordinator / Primary Curriculum Coordinator
Type	Primary Customer
Profile	Responsible for selecting and approving educational technology solutions for the school.
Needs	A reliable, curriculum-aligned, safe, and innovative learning tool that improves student engagement and supports classroom teaching.
Goals	To introduce effective digital learning solutions that enhance educational quality and support modern teaching methods.
Pain Points	Difficulty finding age-appropriate AR tools that are educationally meaningful, easy to implement, and suitable for young learners.
How the System Helps	The AR storybook provides a research-based, child-friendly, and curriculum-relevant learning solution for Grade 1 science education.

Table 6.2: Secondary Customer Persona

Category	**Details**
Persona Name	Grade 1 Science Teacher
Role	Early Primary Science Teacher
Type	Secondary Customer / Direct User
Profile	Uses the system in classroom teaching to help students understand plant-related science concepts in an interactive way.
Needs	A classroom-friendly tool that is easy to use, engaging for children, and supportive of concept understanding.
Goals	To make science learning more interesting, understandable, and memorable for Grade 1 learners.
Pain Points	Traditional teaching methods may not always keep young learners engaged or help them clearly visualize plant-related concepts.
How the System Helps	The AR storybook combines storytelling, visuals, and guided interaction to make science lessons more engaging and easier to understand.

Table 6.3: Indirect Customer Persona

Category	**Details**
Persona Name	Parent of a Grade 1 Learner
Role	Parent / Guardian
Type	Indirect Customer
Profile	Supports the child's learning at home and values meaningful educational experiences beyond the classroom.
Needs	A safe, engaging, and educational tool that helps the child learn effectively in a guided or supervised setting.
Goals	To support the child's curiosity, enjoyment, and understanding of science concepts at home.
Pain Points	Limited access to interactive educational tools that are both enjoyable and academically useful for young children.
How the System Helps	The AR storybook supports emotional engagement, guided learning, and progress-related feedback, making it useful for supervised home learning as well.

6.2 Value Proposition

The main value proposition of the proposed system is that it offers a child-centered, safe, and engaging AR learning experience that combines physical storybooks with interactive digital content to improve understanding of plant-related science concepts. The proposed solution already emphasizes that printed storybook pages, camera input, mobile sensor data, and guided interactions are integrated to create an AR learning experience that is more meaningful than static textbook diagrams. In addition, the system is differentiated by its coordinated research modules, particularly the Hybrid Interaction Framework, which addresses touch accessibility, anchoring between physical pages and 3D content, split-attention reduction, occlusion handling, and fatigue-aware interaction support.

From a commercialization perspective, this gives the product clear value for schools: it is not just an AR storybook, but a research-driven educational tool designed specifically for young learners' usability, comfort, and engagement needs. This makes it more attractive than generic AR applications that may focus mainly on visual novelty without considering children's fine motor skills, physical comfort, or cognitive load.

6.3 Revenue Model

A suitable revenue model for this product is a business-to-school licensing model. Under this model, schools would pay an annual or term-based subscription for access to the AR storybook application, learning content, teacher support features, and software updates. This approach is practical because the system is intended for structured educational use rather than one-time casual entertainment. A second possible model is a per-classroom or per-device licensing structure, where schools pay based on the number of student devices or class groups using the system.

An additional revenue stream could come from content expansion packs, such as new storybooks, additional science topics, or curriculum-aligned activity sets for later grades. Since the proposed architecture is modular and includes multiple coordinated research components, the solution has the potential to be extended beyond the initial plant-growth topic into broader environmental or science learning domains. In the longer term, the product could also be offered through partnerships with educational publishers or private tuition and learning centers.

6.4 Cost Estimation

The main cost categories of commercialization would include software development and refinement, content production, 3D asset and animation development, testing and quality assurance, teacher onboarding materials, and maintenance/support costs. The project requires practical development with Unity 3D, AR Foundation, ARCore, and ARKit, as well as ongoing prototyping, iteration, and testing. These factors suggest that the early commercialization stage would involve costs related to continued technical development, optimization, and device compatibility testing.

There would also be costs for producing high-quality storybook pages, visual assets, and classroom-ready deployment materials. If the product is marketed to schools, teacher guides, setup instructions, and user support documentation would also be necessary. In the pilot stage, costs may be relatively high per user because the product is still being refined. However, once the core application and content pipeline are established, the marginal cost of distributing the software to additional schools would be lower than that of physical-only learning tools.

6.5 Pricing Strategy

The pricing strategy should be based on institutional affordability and educational value. A practical approach would be to offer a school subscription package with different tiers depending on the number of users, classes, or content modules required.

Because the target customers are schools rather than individual mass-market consumers, the pricing should remain reasonable enough for adoption while also reflecting the research-based and curriculum-aligned value of the product. A pilot pricing strategy may also be used in the initial stage, where selected schools receive the system at a discounted rate in exchange for implementation feedback and case-study data. This would support both early adoption and product refinement.

6.6 Competitive Advantage

The proposed solution has several competitive advantages. First, it combines physical storytelling with handheld AR in a way that is explicitly designed for young learners rather than adapted from adult-centered AR systems. The Hybrid Interaction Framework includes predictive UI, dynamic button resizing, physical book-related interaction triggers, and fatigue-related interaction design to improve usability and comfort for children in handheld AR learning. This gives the product a strong advantage over many existing AR learning tools that focus mainly on visual interaction without addressing motor accessibility, ergonomic comfort, or split-attention reduction.

Second, the product is built as an integrated educational ecosystem rather than as a standalone AR animation tool. The complete system includes a Hybrid Interaction Framework, an Affective Visualization Model, a Narrative Engine, and a Context-Aware Simulation Engine working together as coordinated modules. This means the product can offer emotionally responsive visuals, story-driven engagement, and real-world sensor-based interaction in addition to usability improvements. Third, because the solution is intended for early science education and is aligned to a school curriculum context, it has stronger classroom relevance than general-purpose AR entertainment applications.

6.7 Intellectual Property Considerations

The intellectual property (IP) of the proposed product would likely include several protectable elements: the software implementation, the interaction design logic, the storybook content, the 3D assets and animations, and possibly the novel integration of hybrid interaction mechanisms such as adaptive UI resizing, physical anchors, book-related triggers, and fatigue-aware flow design. These aspects may support future claims for software ownership, design protection, or publication-based research recognition.

At the commercialization stage, it will be important to clearly determine the ownership of the software, assets, and content among the student researchers, the university, and any future industry partners. Copyright protection would apply to the storybook design, visuals, animations, and source code, while trademark protection may later be considered for the product name and brand identity. If the final implementation contains

sufficiently novel technical methods, there may also be potential for patent-related discussion, although that would require legal and institutional review. In addition, care must be taken to avoid infringement when using third-party libraries, asset packs, or audio resources during product development.

7. BUDGET AND JUSTIFICATION

The budget for this study has been prepared as a low-cost student research budget by prioritizing free development tools, lightweight local storage, and commonly available mobile devices. Since the project focuses on a handheld AR learning prototype, the required resources are limited to essential hardware, basic testing materials, free or student-access software, minimal cloud support, and practical research-related expenses.

Table 7.1: Estimated Budget for the Research Project

Category	**Description**	**Estimated Cost (LKR)**
Hardware Costs	One mid-range smart tablet for AR testing, if a personal device is not sufficient.	35,000
	Printed storybook pages / markers / prototype book materials for repeated testing.	5,000
	Simple device stand / holder / protective cover for testing sessions.	3,000
Software Licenses	Unity Personal, AR Foundation, ARCore, ARKit, Visual Studio Community. Graphic/audio asset support using free or student-access tools.	Free
Cloud Services	Google Drive / OneDrive free tier for backup, documentation, research files, logs and test data.	3000
Human Effort Cost	Contingency allocation for minor development and testing expenses.	8,000
	Total Estimated Cost	**54,000**

Table 7.2: Budget Justification for the Research Project

Category	**Justification**
Hardware costs	A compatible smart device is essential because the prototype is designed as a handheld AR application. Printed storybook pages and image markers are required to connect the physical book with the AR content. A phone stand helps maintain stable testing conditions, while charging accessories support uninterrupted sessions during demonstrations and user testing.
Software licenses	The project is intentionally designed around free or student-access tools such as Unity Personal, AR Foundation, ARCore, ARKit, and Visual Studio Community. This keeps software cost at zero while still providing all necessary development features for prototype creation.
Cloud services	Cloud cost is free because GitHub Student Developer Pack provides cloud storage, ensuring no additional cost for Cloud Services.
Human effort cost	No salary cost is allocated because this is a student research project. These expenses only cover practical support such as participant refreshments, local travel, and coordination needed for conducting pilot testing and data collection.

8. WORK BREAKDOWN STRUCTURE (WBS)

The proposed research adopts a Work Breakdown Structure (WBS) based on the four core components of the handheld AR storybook system. The system functions through the interaction between printed storybook pages, mobile camera and sensor inputs, and user actions, while its internal architecture is divided into four coordinated modules: the Hybrid Interaction Framework, the Affective Visualization Model, the Gamified Narrative Engine, and the Context-Aware Simulation Engine. This modular arrangement enables the project work to be distributed logically among team members, with each component addressing a distinct functional area while contributing to the overall system. The final deliverables are expected to include the AR storybook application, learning analytics generated through user interaction, and attitude-related outcome indicators. Overall, this structure provides a clear foundation for task allocation, systematic development, and effective integration of the complete solution.

Table 8.1: Task Breakdown per Member

Member	**Component**	**Task Breakdown**
IT22641410 – Wijayarathne K.N.	Hybrid Interaction Framework	Design and implement the dynamic UI mechanism for children's fine motor support; develop physical anchors and occlusion handling for printed storybook interaction; design fatigue-aware interaction flow to reduce prolonged arm strain; conduct usability testing related to cognitive load, engagement, and split-attention reduction
IT22179258 – Palliyaguruge S.N.	Affective Visualization Model	Design AR-optimized 3D plant assets; create animations for plant health, stress, and recovery; apply visual and audio effects to improve emotional readability; conduct child-centered AR trials to refine interpretation and empathy-oriented visual performance.

IT22237804 – Amarawardhane A.S.	Gamified Narrative Engine	Develop the branching story structure; create AI/rule-based dialogue and feedback according to plant condition and context; implement time-delta persistence for plant growth and recovery; design redemption-based rewards and evaluate environmental attitudes and responsibility.
IT22626974 – Dissanayaka D.M.D.D.	Context-Aware Simulation Engine	Develop light-sensing functionality; create algorithms to simulate plant growth based on environmental conditions and user actions; optimize sensor aggregation for efficiency; implement flashlight-assisted growth support and gyroscope-based wind/water simulation.
All Members	Integration and Evaluation	Integrate the four modules into one handheld AR storybook system; perform joint system testing, debugging, user evaluation, refinement, presentation preparation, and final product completion

Table 8.2: Timeline Alignment

Time Period	Planned Activities	Related Deadline
March 2026	Finalize proposal, refine requirements, confirm component scope, and prepare initial designs for all four modules.	Proposal Submission / Proposal Presentation
April 2026	Begin early prototype development for interaction design, visualization, narrative logic, and simulation logic.	Proposal milestone continuation
May 2026	Complete first-stage implementation of core module functions and prepare for initial review.	Progress Presentation 1 / Check List Submission
June 2026	Continue implementation and begin cross-component integration, including AR page flow, module communication, and output handling.	Post-Progress 1 development phase
July 2026	Conduct usability testing, debugging, integration refinement, and preliminary evaluation of the overall prototype.	Preparation for Progress Presentation 2
August 2026	Complete the full product, finalize working prototype, and ensure all four modules function together as one integrated system	Progress Presentation 2
September 2026	Improve stability, finalize evaluation, organize documentation, and prepare report and website content.	Final documentation phase
October 2026	Submit final report, website, final checklist, and complete final presentation and viva.	Final submission milestones
December 2026	Prepare and submit research paper publication evidence	Research paper publication evidence submission

Table 8.3: Balanced Workload Distribution

Work Area	**Workload Distribution Explanation**
Component Ownership	Each member is assigned one major research component, which creates clear responsibility and avoids unnecessary overlap. This supports balanced distribution across interaction, visualization, narrative, and simulation development.
Technical Development	The work is balanced because all four members develop a comparable subsystem with its own design, implementation, and testing responsibilities.
Integration Work	Integration is shared because the final system depends on interaction among all four modules inside one AR storybook platform. This includes combining sensor input, page tracking, story flow, visualization, and usability support.
Testing and Evaluation	Evaluation is also shared, since the project collects observation data, interaction logs, plant growth status, and pre/post outcomes, and requires ethical handling of child participants.
Final Deliverables	Report writing, presentation preparation, website completion, and final refinement are group responsibilities, ensuring that no single member carries all final-stage work.

9. GANTT CHART

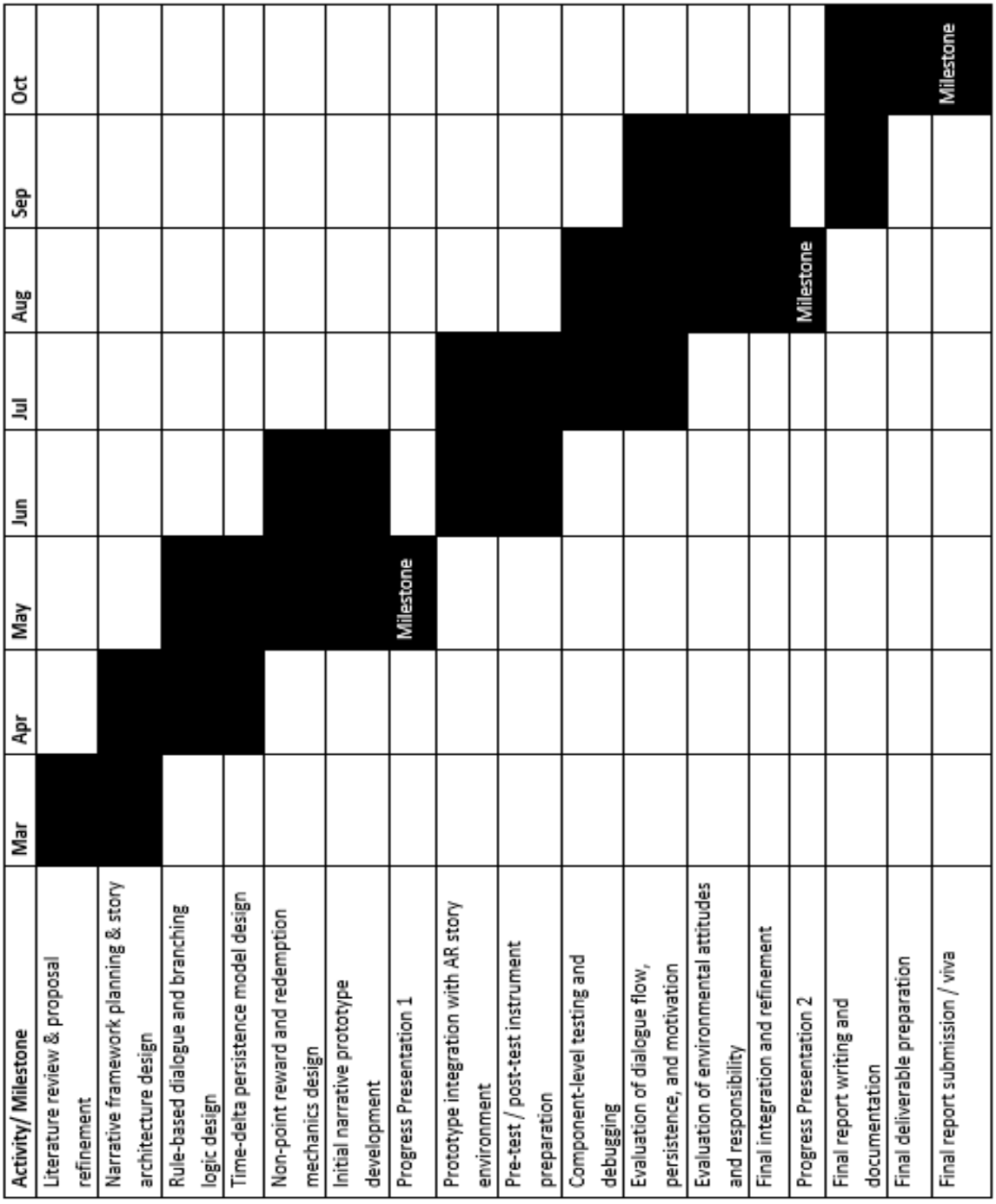


Figure 3: Gantt Chart

This Gantt chart shows that the proposed **gamified narrative engine** begins with requirement analysis, literature review, and research-gap refinement during the proposal stage. The development phase then focuses on four key technical areas: **rule-based narrative generation, time-delta persistence, meaningful gamification**

design, and redemption-based digital storytelling. The middle phase of the schedule is allocated for prototype development, AR integration, and testing, while the later phase is dedicated to evaluation of narrative flow, learner motivation, environmental attitudes, refinement of the component, documentation, and final submission. This timeline ensures that the component is developed in a systematic manner, remains aligned with the overall research project schedule, and is ready for integration into the complete AR storybook system before the final presentation.

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11.APPENDICES

11.1 Appendix A – Survey Instruments

This appendix presents the sample instruments that will be used to collect data during the evaluation of the proposed gamified narrative framework in the handheld AR storybook system. Since the study involves young learners, direct questionnaires will be kept simple and supported by teacher or facilitator observation. The main instruments will include a structured observation sheet, a teacher/facilitator feedback form, and a simple pre-test / post-test evaluation format related to plant care, environmental responsibility, and learner response to the narrative experience.

A.1 Observation Sheet

The observation sheet will be used during learner interaction sessions to record behavior related to narrative understanding, engagement, responsibility, and response to story consequences.

Table 11.1: Observation sheet

Observation Area	**Description**
Narrative Understanding	Whether the child understands the story flow and follows the sequence of events correctly
Choice Response	Whether the child understands that their choices affect the plant and story outcome
Engagement Level	Whether the child remains interested and attentive during the narrative activity
Emotional Response	Whether the child shows concern, empathy, or positive reaction toward the plant's condition
Redemption Behavior	Whether the child attempts to correct mistakes when redemption options are provided
Task Completion	Whether the child completes the narrative activity successfully

A.2 Teacher / Facilitator Feedback Form

The teacher or facilitator will be asked to evaluate the clarity, educational value, and child suitability of the narrative framework.

Sample statements:

- The story structure was understandable for the learner.
- The child was able to connect their choices with plant outcomes.
- The dialogue and feedback were appropriate for the learner's age.
- The persistence and recovery behavior helped the learner understand plant care over time.
- The narrative experience supported environmental learning and engagement effectively.

A.3 Pre-Test / Post-Test Evaluation

This will include a short teacher-assisted evaluation of learner understanding before and after the session, focusing on:

- understanding of plant needs and care
- awareness of how actions affect the plant
- interest in helping the plant recover
- feelings of responsibility toward living things
- willingness to repeat the activity

11.2 Appendix B – Detailed Diagrams

This appendix contains the detailed technical diagrams used to explain the structure and operation of the proposed system and the selected component.

The appendix may include:

- Overall System Architecture Diagram
- Gamified Narrative Framework Architecture Diagram
- Narrative Flow / Branching Diagram
- Persistence and Reward Logic Diagram
- Gantt Chart / Work Plan Diagram

B.1 Overall System Architecture

This diagram presents the complete AR storybook system with the four research components:

1. Hybrid Interaction Framework
2. Affective Visualization Model
3. Gamified Narrative Engine
4. Context-Aware Simulation Engine

B.2 Individual Component Diagram

This diagram shows the internal structure of the **Gamified Narrative Engine**, including:

- Rule-Based Narrative Generation Module
- Dialogue and Feedback Module
- Time-Delta Persistence Module
- Gamification and Intrinsic Motivation Module
- Redemption and Restoration Module
- Attitude / Responsibility Evaluation Support Module

B.3 Data Flow Diagram

This diagram shows how learner choices move through the system, including:

- learner input
- narrative rule checking
- plant-state update
- reward or redemption trigger
- dialogue response
- updated narrative state and logging

11.3 Appendix C – Preliminary UI Mockups

This appendix includes the early visual designs of the proposed handheld AR storybook system. These mockups are intended to demonstrate the planned narrative interaction style and user interface structure before full implementation.

C.1 Planned UI Design Features

The preliminary UI is expected to include:

- simple dialogue boxes
- large story-choice buttons
- plant mood and health indicators
- visual prompts for care actions
- minimal text with child-friendly wording
- guided feedback after each narrative decision

C.2 Purpose of Mockups

These mockups are included to show how the interface design supports:

- easier understanding of story choices
- clear link between action and plant outcome
- guided redemption actions
- reduced confusion during narrative interaction
- stronger learner focuses on plant care and consequence

C.3 Example Screens

The appendix may include:

- Story Start Screen
- Plant Interaction Screen
- Choice Selection Screen
- Redemption / Recovery Screen
- Post-Activity Reflection Screen

11.4 Appendix D – Risk Analysis

This appendix presents the main risks associated with the research and the proposed mitigation strategies.

Table 11.2: Risk Analysis

Risk	**Description**	**Mitigation Strategy**
Complex Narrative Flow	Learners may become confused if too many branches or outcomes are introduced	Keep story paths simple, use child-friendly dialogue, and pilot test narrative flow
Inconsistent Dialogue Response	Rule-based feedback may not always match learner context properly	Refine rule conditions through repeated testing and expert review
Persistence Errors	Plant growth, decay, or recovery may not update correctly across inactive sessions	Validate time-delta calculations and test multiple session scenarios
Weak Motivation	Learners may not respond strongly to non-point rewards	Strengthen ecological feedback, emotional cues, and visible consequences
Redemption Misunderstanding	Children may not understand how to correct earlier mistakes	Use guided prompts and explicit recovery actions in the narrative
Limited Learner Access	Difficulty in arranging child-centered pilot sessions	Coordinate early with schools and maintain alternative testing schedules
Data Collection Delays	Observation sheets, feedback forms, or evaluation data may not be completed on time	Prepare instruments in advance and follow a structured evaluation plan
Ethical Concerns	Child-related data must be handled carefully	Use consent, assent, anonymization, and secure academic data handling procedures

11.5 Appendix E – Ethical Documents

This appendix includes the ethical and participant-related documents required for the research study, especially because the target participants are young learners.

The appendix may include:

- Parent / Guardian Consent Form
- Child Assent Form
- Participant Information Sheet
- Privacy and Confidentiality Statement
- Ethical Clearance Submission Documents

E.1 Parent / Guardian Consent Form

This document will explain:

- the purpose of the study
- the child's involvement
- the type of AR narrative activities to be carried out
- the voluntary nature of participation
- privacy and confidentiality protection measures

E.2 Child Assent Form

This will provide a simple explanation, in age-appropriate language, to confirm that the child is willing to participate in the AR storybook activity.

E.3 Privacy and Confidentiality

All collected data will be anonymized. No unnecessary personal information will be stored, and all research data will be used only for academic purposes. Interaction logs, observation records, and pre-test / post-test results will be handled securely and accessed only by authorized academic personnel.